[Abstract]

LoRA: Low-Rank Adaptation of Large Language Models

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Abstract: An important paradigm of natural language processing consists of large-scale pre-training on general domain data and adaptation to particular tasks or domains. As we pre-train larger models, full fine-tuning, which retrains all model parameters, becomes less feasible. Using GPT-3 175B as an example, deploying independent instances of fine-tuned models, each with 175B parameters, is prohibitively expensive.

We propose Low-Rank Adaptation, or LoRA, which freezes the pre-trained model weights and injects trainable rank decomposition matrices into each layer of the Transformer architecture, greatly reducing the number of trainable parameters for downstream tasks. Compared to GPT-3 175B fine-tuned with Adam, LoRA can reduce the number of trainable parameters by 10,000 times and the GPU memory requirement by 3-fold.

[Low-Rank Parameterized Update Matrices]

4.1 Low-Rank Parameterized Update Matrices: A neural network contains many dense layers which perform matrix multiplication. The weight matrices in these layers typically have full-rank. When adapting to a specific task, pre-trained language models have a low intrinsic dimension and can still learn efficiently despite a random projection to a smaller subspace.

For a pre-trained weight matrix W_0 in R^(d x k), we constrain its update by representing the latter with a low-rank decomposition W_0 + Delta W = W_0 + B A, where B in R^(d x r), A in R^(r x k), and the rank r << min(d, k). During training, W_0 is frozen and does not receive gradient updates, while A and B contain trainable parameters.

We use a random Gaussian initialization for A and zero for B, so Delta W = B A is zero at the beginning of training. Unlike Adapters, LoRA introduces zero additional inference latency because the trainable matrices B A can be explicitly merged with the frozen weights W_0 during deployment.

[Empirical Evaluation on GLUE and WikiSQL]

5 Empirical Experiments: We evaluate the downstream task performance of LoRA on RoBERTa, DeBERTa, GPT-2, and GPT-3 175B across the GLUE benchmark, WikiSQL, and SAMSum datasets. In the Transformer architecture, we follow Vaswani et al. (2017) and apply LoRA to the Self-Attention projection matrices W_q and W_v.

On GPT-3 175B, LoRA matches or exceeds the full fine-tuning baseline on WikiSQL (achieving 74.0% accuracy) and MultiNLI (achieving 91.7% accuracy) while updating only 4.7M parameters compared to 175,255M parameters in full fine-tuning, reducing VRAM memory consumption from 1.2 TB to 350 GB.